

Macroeconomic Theory and Policy: Lecture W9 (Ch.15)

What are You Going to Learn (Ch. 15)

- Introducing DSGE model, which
 - ▶ incorporate microfoundations, dynamics, general equilibrium, and shocks
 - ▶ predict how the economy evolves over time in response to shocks

A Brief History of DSGE Models

- Robert Lucas, Finn Kydland, Ed Prescott, Tom Sargent, and Chris Sims revolutionized modern short-run macro models in the 1970s and 1980s
- DSGE referred to as dynamic, stochastic general equilibrium (DSGE) models
 - ▶ Dynamic: examine the economy over time
 - ▶ Stochastic: involve shocks that lead to macroeconomic fluctuations
 - ▶ General Equilibrium: multiple markets clear simultaneously
 - ★ Labour markets, output market, and other markets endogenously determine all prices

A Brief History of DSGE Models

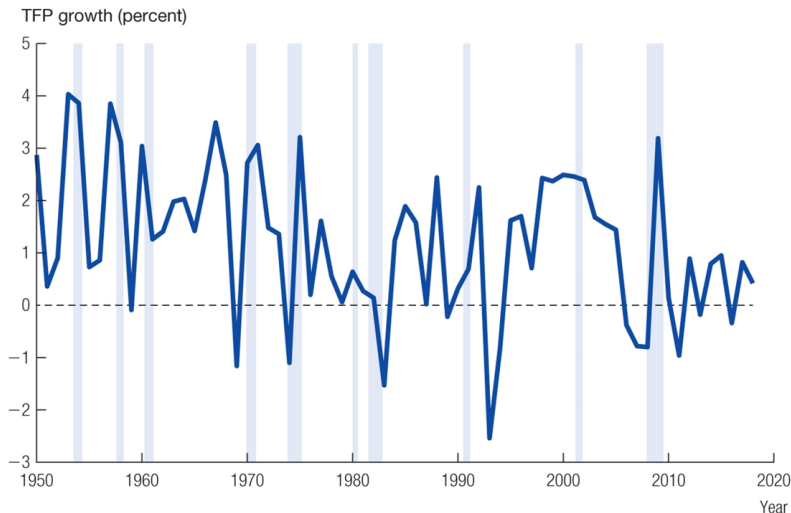
- The first popular application is the Real business cycle models
- Used Solow growth model to study macroeconomic fluctuations
 - ▶ Assume TFP changes over time, instead of growing at a constant rate or being a parameter
 - ▶ Add consumer and firm maximization problem (adding microfoundation)

A Brief History of DSGE Models

- Introduced total factor productivity (TFP) shocks
 - ▶ Positive shock: new technology, institutions (e.g., introduction of the Internet)
 - ▶ Negative shock: institutions, taxes (e.g., Changes in taxes)
- TFP growth from the Solow growth accounting formula shows large fluctuations in TFP
 - ▶ However, as the model captures more feature of the economy, the TFP explain smaller proportion of the fluctuations

U.S. Total Factor Productivity Growth

- The movements in TFP seems important in explaining business cycles



Source: John Fernald, "A Quarterly, Utilization-Adjusted Series on Total Factor Productivity," Federal Reserve Bank of San Francisco, www.frbsf.org/economic-research/indicators-data/total-factor-productivity-tfp/.
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A Brief History of DSGE Models

- TFP shocks:
 - ▶ Positive shocks cause the economy to boom
 - ▶ Negative shocks cause GDP to decline
- The economy would go through cycles because of TFP shocks
- Shocks to TFP are “real” in RBC and cause output to fluctuate as in business cycle
- Some weaknesses:
 - ▶ Not all macroeconomic fluctuations are driven by TFP shocks (e.g., 2001)

From Real Business Cycle to DSGE

- Nowadays, DSGE models have moved away from the “real” classification by incorporating monetary policy
- Components of modern DSGE models
 - ▶ Endogenous variables
 - ▶ Shocks
 - ▶ Features that affect the way shocks impact endogenous variables over time, for example,
 - ★ Various tax (capital, income, etc)
 - ★ Capital adjustment (firms need time to adjust the level of capital)
 - ★ Unemployment insurance

Endogenous Variables

- DSGE models can be large or small systems
- One important component of almost all DSGE models is the labour market. Decisions in the labour market determine how much output is produced:
 - ▶ From the unemployment model we studied, if we want a rich labor market model, we will need Hours worked, Searching for employment, Job effort
- Below is a list of potential variables to be included in DSGE:
 - ▶ wages, interest rates, inflation, hours worked, GDP, consumption, investment, government purchases, employment, unemployment, capital, international trade, foreign debt, government debt, exchange rates

Shocks

- Shocks cause endogenous variables to fluctuate
 - ▶ Include total factor productivity, government purchases, taxes, monetary policy, energy prices, and financial frictions
- Shocks can be temporary or permanent
 - ▶ Some shocks may last only one period. Some shocks may last forever
 - ▶ For instance, government spending may be temporarily high this year, or there could be a long-lasting change in the size of the government

Shocks

- Shocks may be current or anticipated
 - ▶ A shock may occur now or instead we may get news today of a shock that will occur in the future
 - ▶ For example, people anticipate AI would enhance productivity
 - ▶ When news arrive, agent would integrate the news with rational expectation in the decision making
 - ▶ Uncertainty about the future can slow economic activity today as well

Features and Mathematics

- DSGE models are complex to solve mathematically because they involve many individual decisions
 - ▶ DSGE models is calculated using numerical simulations most of the time
- Some of the DSGE model's common features:
 - ▶ Nominal rigidities
 - ★ Monetary policy only has real effects if some friction interferes with price adjustment
 - ★ e.g., Sticky nominal wages and sticky prices

Features and Mathematics

Some of the DSGE model common features:

- Adjustment costs:
 - ▶ It is expensive to change the capital stock by a large amount
 - ▶ The dynamic response of variables to shocks is slow to adjust
 - ▶ The effect of the shock occurs several periods after the shock, the opposite of the Solow model, where the effect of the shock is the largest immediately after it hits

Features and Mathematics

Some of the DSGE model common features:

- Incomplete markets:
 - ▶ Incomplete markets refer to situations where certain financial instruments or contracts are not available to fully hedge against all risks or uncertainties in the economy (i.e., just like in real life)
 - ▶ The early version of economics model assume agent can fully hedge
 - ▶ In such markets, individuals may constraints to inefficiencies and potentially suboptimal outcomes
 - ▶ Incomplete markets can arise due to factors such as adverse selection, moral hazard, transaction costs, regulatory constraints, or the presence of non-tradable assets, time or geography

Features and Mathematics

Some of the DSGE model common features:

- Heterogeneity: People and firms are different and response to shocks differently
 - ▶ Some individuals have completely inelastic labour supply, work 40 hours per week or not at all. Meanwhile, some may have the average elastic or very elastic
 - ▶ Also Reservation wage may different across population
 - ▶ When shocks change the wage, people enter and exit from the labour market
 - ▶ The aggregate elasticity depends on the distribution of reservation wages rather than on any individual's elasticity
 - ▶ When analyzing policy impacts on the population, it would be important to understand which demographic groups will be affected by the policy
 - ▶ The differences between agent can be based on various thing, for example, innate ability, wealth, income, etc

A Stylized Approach to DSGE

- Most DSGE models contain at least three markets: Labour, Capital and Output
- Focus on the labour market here
- labour demand:
 - ▶ Derived from firm's profit-maximization problem
 - ▶ Decide how much labour to hire
 - ▶ Negative slope—diminishing MPL
- labour supply:
 - ▶ Decisions by people about how much to work

A Stylized Approach to DSGE: labour Demand

- The labour demand curve is derived from the firm's profit maximization problem
- Firms hire labour until the MPL equals the wage
 - ▶ The MPL is dictated by the production function: $Y = AN^\alpha K^{1-\alpha}$
 - ▶ Marginal product of labour: the amount of extra output produced by the last worker hired
- Hire until $MPL = \alpha \frac{AN^\alpha K^{1-\alpha}}{N} = w$
- Diminishing marginal returns to labour (exponent less than one) implies that each additional unit of labour that is hired produces a smaller and smaller gain in output
- Because of the assumption that the MPL declines as more labour is hired, the labour demand curve is downward sloping
- Note here that the TFP shocks (A) can directly affecting the MPL

A Stylized Approach to DSGE: labour Supply

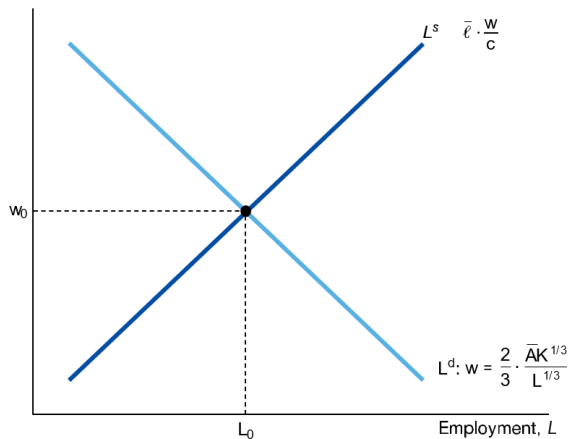
- labour supply is derived from people utility maximization problem with consumption/leisure choice
- Use the following specification: $L^S = \bar{\ell} \frac{w}{c}$
where w is the real wage, c denotes consumption per person and $\bar{\ell}$ is the parameter that governs the overall magnitude of labour supply
- labour supply is increasing in the wage and depends on the wage-to-consumption ratio
- Over time, the real wage increases, and consumption per person also rises
 - ▶ In fact, these growth rates are approximately the same, so that w/c is relatively stable
- Therefore, the amount of labour supplied (per person) would be stable as well

The labour Market in the DSGE Model

- The equilibrium determines the real wage and the amount of labour that gets used in production

The Labor Market in the DSGE Model

Real wage, w



Using the Stylized DSGE Model

Examine the following:

- a TFP shock
- a change in taxes
- a rise in the size of government purchases
- a shock to monetary policy

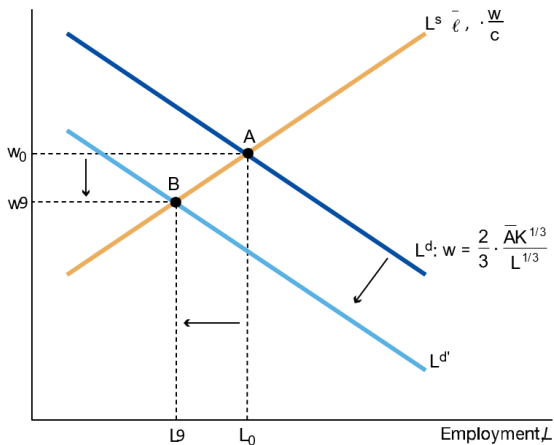
A Negative TFP Shock in the DSGE Model

- A temporarily negative shock that reduces the TFP parameter
 - ▶ Lower A decreases the amount of output that an extra worker can produce.
 - ▶ The MPL is lower for any given L
- Firms do not want to produce due to low demand, so labour demand shifts down
- The TFP shock is temporary, so effects on consumption are small (i.e., no shift on labour supply curve)
- In response to a negative TFP shock, employment falls, and the equilibrium wage falls
 - ▶ A nice result that would match business cycle fact, lead the early DSGE models to focus on TFP shocks

A Negative TFP Shock in the DSGE Model

A Negative TFP Shock in the DSGE Model

Real wage, w



The Imposition of a Sales Tax in the DSGE Model

- Suppose firms must pay a tax for every dollar of output they sell temporarily
- Since firms pay the tax, the labour demand curve shifts
 - ▶ This reduces the amount that the firm earns at any given level of employment, lowering the MPL and shifting the labour demand curve in
- Firms now hire until the after-tax marginal product of labour falls to equal the wage

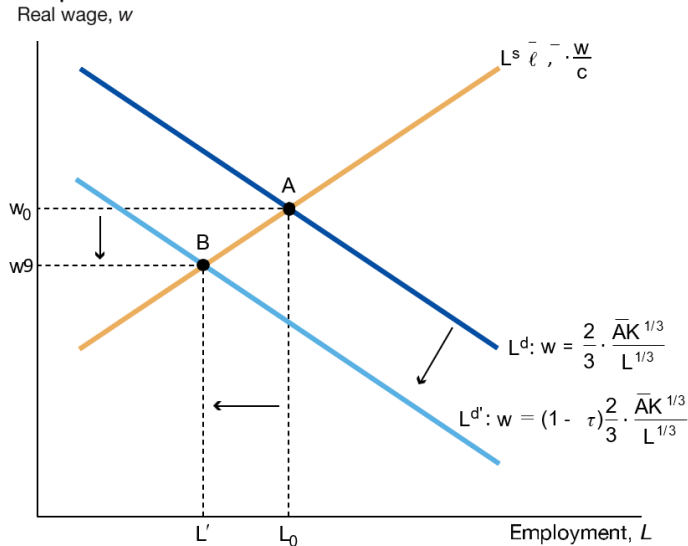
$$(1 - t)\alpha \frac{Y}{N} = w$$

- There are no effect on consumption. Once again, no shift on the labour supply curve

The Imposition of a Sales Tax in the DSGE Model

- Wages and employment decline

The Imposition of a Sales Tax in the DSGE Model

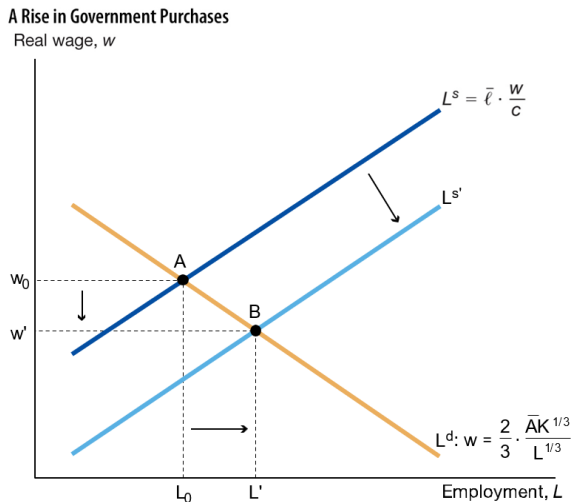


A Rise in Government Purchases

- Consider an increase in government purchases, financed through a rise in lump-sum taxes at some unspecified point in the future
- The implied tax burden lowers permanent income and therefore reduces consumption today
 - ▶ Consumption is the variable in our DSGE model that allows future events to affect the economy today
- A general principle in microeconomics is that as people get richer, they generally consume more of all goods, including leisure
 - ▶ This means that consumption and leisure will typically move in the same direction
 - ▶ This means the labour supply curve shifts to the right
- There are no change to labour demand curve this time

A Rise in Government Purchases

- The equilibrium effect of this shock is to increase employment and reduce the real wage



Sticky Wages

- For monetary policy to have real effects on the economy, we must introduce frictions in the adjustment of prices
 - ▶ If the classical dichotomy holds, changes in monetary policy does not have any effect on real variables
- Distinguish between nominal and real variables
 - ▶ W is the nominal wage
 - ▶ P is the price level
 - ▶ $W/P = w$ is the real wage

Sticky Wages in the DSGE Model

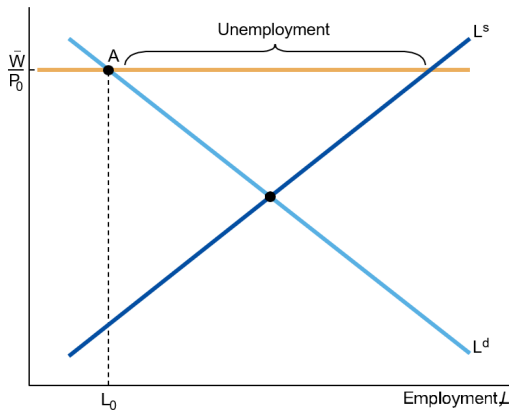
- Under sticky wages, the wage does not adjust when the market condition change

$$W = \bar{W}$$

- Employment is set by the demand side in this case, as both sides of the trade must be willing participants – take the one with lower employment level

Sticky Wages in the DSGE Model

Real wage, W/P



A Monetary Expansion with Sticky Wages

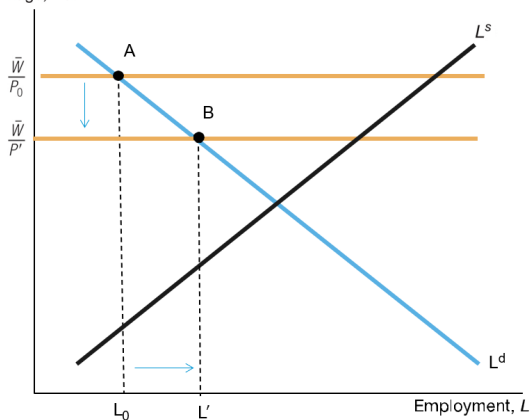
- When there is expansionary monetary policy, lowering the federal funds rate increases other prices (like in our AD curve)
 - ▶ But wages are still sticky
- The overall price level P rises to some new level P'
- With the nominal wage stuck at W , the price increase leads the real wage W/P to decline
- The equilibrium is then move down along the demand curve

A Monetary Expansion with Sticky Wages

- At this lower real wage, labour demand is higher and labour supply is lower
- This is not the signature pattern we would expect. Real wages are generally low during Booms

A Monetary Expansion with Sticky Wages

Real wage, W/P

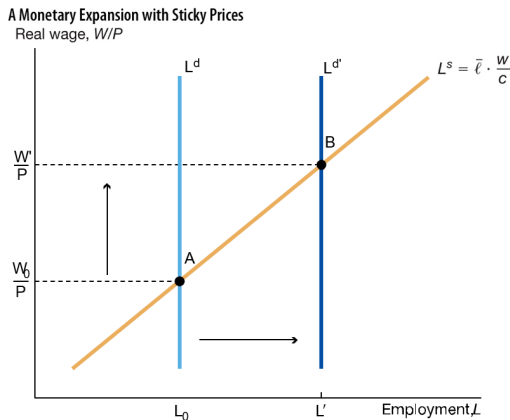


A Monetary Expansion with Sticky Prices

- In the Keynesian tradition, the output is determined by the demand
 - ▶ Firms must demand the amount of labour that is needed to produce the demanded output
 - ▶ In this case, the labor demand curve is now vertical
- Consider the effect of an expansionary monetary policy (a lower fed funds rate) under sticky prices (i.e., $P = \bar{P}$ instead of wages)
- The monetary expansion increases demand in the economy since prices are fixed
- Firms need to produce a higher level of output and therefore require more labor: the labor demand curve shifts to the right

A Monetary Expansion with Sticky Prices

- To induce more people to work, the wage must rise (recall that the nominal wage is flexible in this case), and with sticky prices, this leads to an increase in real wages, as the economy moves to point B



Next Lecture

- Next lecture will finish Ch. 15 and begin Ch. 18